Standing Waves on a String: Measuring g From a Vibrating String

SYNOPSIS

A string fixed at both ends will only stand still in certain shapes, and which shapes it takes depends on how hard it is pulled. You will drive a string at its third harmonic for ten different hanging masses, and from the straight line that comes out of that data you will recover a constant you already know the value of — the acceleration due to gravity. Recovering a known constant is how you test whether an equation is right.

Code LB2-M1 **Due** See the Module 1 Lab folder on Brightspace **Pairs with** Note 1 and Note 8

Where Data from the lab video **Report** Full lab report — use the Lab Template

This experiment is filmed in the physics laboratory. **You take your data from the video**, then do the analysis and write the report. Watch it once through before you start recording numbers.

What is in the apparatus

- Pasco wave driver, with a function generator and two plug-type connectors
- String, about two metres
- Lab stand, hook and string attachment
- Pulley with table clamp
- Mass hanger and slotted masses
- Metre stick and electronic lab balance

The function generator produces a sinusoidal voltage whose frequency and amplitude you can set. The wave driver is a strong speaker with an arm on it, and the arm shakes the string.

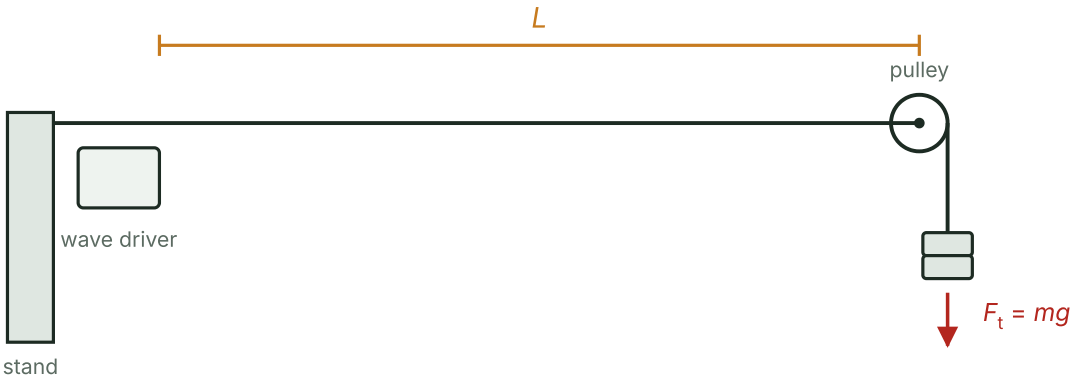


Figure 1. One end of the string is tied to the stand and passes through the wave driver; it runs a length L to the pulley and hangs over it. ***L* is measured between the wave driver and the pulley** — not the whole string. The hanging mass sets the tension: $F_t = mg$.

Theory

The speed of a wave on a stretched string is set by the tension it is under and how heavy it is per unit length:

$$v = \sqrt{(F_t/\mu)} \quad (1)$$

with v in m/s, F_t in N, and the linear density

$$\mu = m_s/L_s \quad (2)$$

where m_s is the string's mass in kg and L_s its **total** length in m. Note that μ uses the whole string, while the tension length L is only the active span.

A string fixed at both ends must have a node at each end, so only certain wavelengths fit. For the fundamental the wavelength is twice the string length, and combining that with $v = f\lambda$ and equation (1):

$$f = (1/2L)\sqrt{(F_t/\mu)} \quad (3)$$

and the higher harmonics are whole-number multiples of it:

$$f_n = n\,f, \quad n = 1, 2, 3, \ldots \quad (4)$$

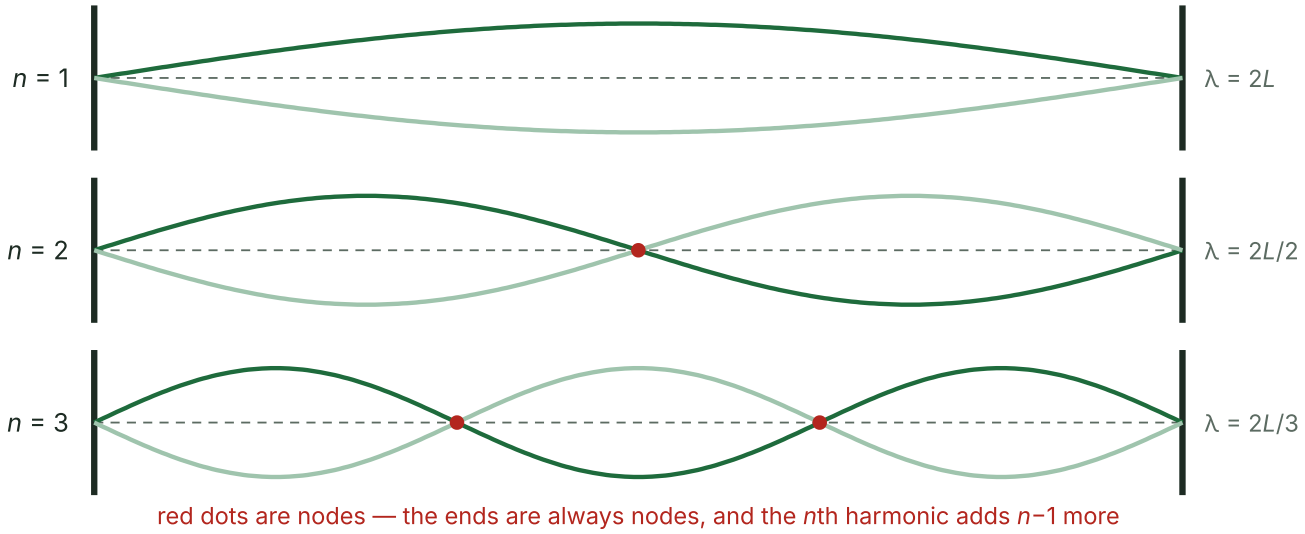


Figure 2. The first three harmonics. Each is drawn from $\sin(n\pi x/L)$, so the loop counts are the equation rather than a sketch. **You will be driving the third** — three loops, two nodes between the ends.

Now substitute $F_t = mg$ into equation (3) and rearrange:

$$f^2 = \left(g \,/\, 4L^2\mu \right) m \quad (5)$$

Why this one is worth doing before you take data

Equation (5) is $y = Ax$, a straight line through the origin, with $y = f^2$ and $x = m$. Everything else — g, L, μ — is bundled into the slope. So g is not something you measure directly in this lab; it is **hidden inside the slope**, and you get it out by comparing the slope you measured against the form of equation (5). **You will derive equation (5) from (1) and (3) in your report**, so do it now, on paper, before the numbers exist to be fitted.

Procedure

1. Make a constants table like Figure 3: total string length L_s, active length L, total string mass m_s, and the mass of the hanger.
2. The string is about two metres. Record its total length in cm.

3. Record the string's mass m_s in g from the balance.
4. Compute the linear density μ in kg/m. **Convert first** — mass to kg, length to m — then divide. Put it in the constants table.
5. The string runs from the stand over the pulley at the bench edge, about 1.5 m.
6. Measure L exactly between the pulley and the wave driver, in m.
7. The hanger is 50.0 g.
8. A 200 g slotted mass brings the total hanging mass to 250 g. Treat these as three significant figures.
9. With the amplitude moderate, raise the frequency slowly until the fundamental appears.
10. Keep going until you see two loops, then three.
11. At the third harmonic, keep raising in 1 Hz steps until the pattern goes unstable, then come back down in tenths until it is clean and steady. **Record that frequency to a tenth of a hertz** as f_3.
12. Make a raw table like Figure 4 for hanging mass m [g] and third-harmonic frequency f_3 [Hz]. Enter the 250 g run as trial 1.
13. Standby the driver, add 50 g, and repeat. Continue in 50 g steps up to 700 g total — **ten trials**.
14. Make a processed table like Figure 5: trial, m [kg], the fundamental f [Hz], and f^2 [Hz²]. The fundamental is the measured third harmonic divided by three, from equation (4).
15. Plot f^2 against m [kg] as a scatter plot. Add a linear trendline, display its equation, and show the R^2 value.
16. From the slope and equation (5), calculate g. Call it g_e.
17. Report the percent accuracy error against the accepted value:

$$(\text{a.e.})_p = |g_e - g| \,/\, g \times 100, \quad g = 9.81\ \text{m/s}^2 \quad (6)$$

Quantity	Symbol	Value	Unit
Total string length	L_s		m
Active length, driver to pulley	L		m
Total string mass	m_s		kg
Linear density	μ		kg/m
Mass hanger		50.0	g

Figure 3. Constants.

Trial	m [g]	f_3 [Hz]
1	250	
2	300	
⋮	⋮	
10	700	

Figure 4. Raw data.

Trial	m [kg]	f [Hz]	f^2 [Hz^2]
1			
⋮			
10			

Figure 5. Processed data.

What your report must contain

1. **Introduction** — goal and procedure, 200 ± 20 words, with an image or screenshot of the setup *and* a sketch of your own with the important parts labelled.
2. **Experimental Data** — the raw tables only (Figures 3 and 4). No processed data here.
3. **Analysis** — sample calculations in order: linear density μ; the fundamental f and its square; g_e; the percent accuracy error. Then the graph, the processed table (Figure 5), and **the algebraic derivation of equation (5) from (1) and (3)**.
4. **Results and Discussion** — g_e, its percent accuracy error, and what you make of them, naming the likely sources of error.
5. **Conclusion** — what you set out to find, what you found, and the percent accuracy error.

> **Make the numbers findable.** Results go on their own line, in bold, as equations with units, not inside a paragraph.

Questions for your discussion

I. Start from a small segment of string under tension F carrying a pulse at speed v, and show how one arrives at equation (1). Show every step.

II. For a wave $y(x,t) = A\sin(kx - \omega t)$ with $A = 2$ cm, find the vertical position y, the vertical speed v_y and the vertical acceleration a_y at $x = 15$ cm and $t = 3$ s, for **trial 5** of your own data. Everything else in the equation comes from your measurements. Differentiate to get v_y and a_y.

> That equation describes a wave travelling *one way*. The real string carries the reflected wave too, so this calculation does **not** predict what the standing wave does — you are deliberately working the one-way case. Say so in your answer.

III. Your μ uses the string's whole length but L uses only the driven span. Why is that right, and what would go wrong if you used the same length for both?

Answer three of these fully rather than all of them briefly. Name the three you chose.

WATCH · THE LAB, FILMED

Recording to come. This is where your data comes from, so watch it before you build your tables.

Files for this lab

HANDOUT · PDF

This lab, printable

have it open beside the video.

EXCEL

Data workbook

constants, raw and processed tables, with the plot set up.

WORD

Report starter

section headings with an opening sentence for each.

WORD

Grading rubric

read it before you write.

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